Factorials can be generalised as double factorial as follows:

Double Factorial of n:

Factorial of an integer n, denoted by $n!$ can be viewed as a function $f : \mathbb{N} \cup \{0\} \to \mathbb{N}$, where $\mathbb{N}$ is the set of all Natural numbers, define as

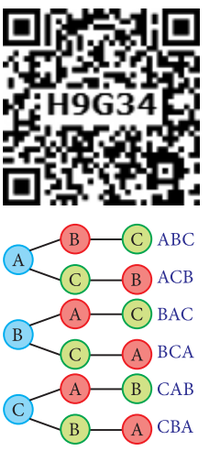
$$f(n) = \begin{cases} 1 & \text{for } n = 0, \\ n \times (n-1) \times (n-2) \times .... \times 3 \times 2 \times 1 & \text{for } n \neq 0. \end{cases}$$

One can defin $n!!$ (double factorial of n) as

$$g(n) = \begin{cases} 1 & \text{for } n = 0, \\ n \times (n-2) \times (n-4) \times .... \times 4 \times 2 & \text{for n is even} \\ n \times (n-2) \times (n-4) \times .... \times 3 \times 1 & \text{for n is odd} \end{cases}$$

Accordingly, $5!! = 5 \times 3 \times 1 = 15$ and $8!! = 8 \times 6 \times 4 \times 2 = 384$.
Note that $n!! \neq (n!)!$ as $4!! = 8$ where as $(4!)! = (24)!$

4.4 Permutations



What is a permutation ?

Permuations come in various disguises.

Suppose three friends A, B and C have to stand in line for a photograph. In how many order can they stand? Some of the possible arrangements (from left to right) are

ABC, ACB, BAC, BCA, CAB, CBA

Thus there are six possible ways in which they can arrange themselves for the photograph.

Thus if 3 objects have to be arranged in a row there are $3 \times 2 \times 1 = 3!$ possible permutations. The number of permutations of 4 objects taken all at a time is $4 \times 3 \times 2 \times 1 = 4!$ Thus if n objects have to be arranged in a line there are $n \times (n-1) \times (n-2) \times \cdots \times 3 \times 2 \times 1 = n!$ possible arrangements or permutations.

Suppose you have 7 letters A,B,C,D,E,F and G. We want to make a 4 letter string. We have 7 choices for the 1st letter. Having chosen the firs 1etter, we have 6 choices for the second letter. Proceeding this way, we have 4 choices for the 4th letter.
Hence, the number of permutations of 4 letter strings chosen from 7 letters is

$$7 \times 6 \times 5 \times 4 = \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1} = \frac{7!}{3!} = \frac{7!}{(7-4)!}.$$

More generally, the number of distinct permutations of r objects which can be made from n distinct objects is $\frac{n!}{(n-r)!}$. It is denoted by ${}^nP_r$. The formal proof of this result will be proved in this section.